Aggregate trace heights of elliptic multiples

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Abstract

We prove a linear height comparison for the sum of the traces of abscissas of nonzero multiples of the fixed elliptic point, and a second comparison for the trace of their product when the labels are odd. The main geometric argument excludes all positive-dimensional abelian translates through the specified points in a trace fiber, including translates defined only over an algebraic extension. A theorem of Von Buhren then supplies a linear comparison with an effective geometric slope and an ineffective additive constant. Under the arithmetic premises of the specified rational source, this proves the existence of a defining polynomial with twenty-seven total variables. The fixed coefficient needed to dominate the additive constant is not determined. The result is therefore an existence theorem, distinct from an explicit polynomial specification.

1 Functions with one torsion zero and one pole

Throughout the geometric argument the base field is algebraically closed of characteristic zero. Let E be an elliptic curve, let O be its identity, and write $\mathcal{R} = \operatorname{End}(E)$. This ring is commutative and has no zero divisors. Let $T \in E$ have finite order $N > 1$. Suppose $f \in k(E)$ satisfies

$$\operatorname{div}(f) = m[T] - m[O] \qquad (m > 0).$$

Thus f has its only zero at T and its only pole at O. Such a function exists when $m = N$, because $N(T - O)$ is a principal divisor on an elliptic curve.

Lemma 1. *Let $P \in E$ have infinite order. Let $n_1, \dots, n_r$ be nonzero integers prime to N, and let $\alpha_1, \dots, \alpha_r \in \mathcal{R}$. At every point U of E, a pole of one function $f([n_i]P + \alpha_i U)$ cannot coincide with a zero of another. Consequently, if some $\alpha_i \neq 0$, the product of these functions has a pole and is nonconstant.*

Proof. Suppose

$$[n_i]P + \alpha_i U = O, \qquad [n_j]P + \alpha_j U = T.$$

The first equation requires $\alpha_i \neq 0$; a constant factor cannot be a zero or a pole because its argument is nontorsion. Applying the two endomorphisms and subtracting gives

$$(n_i\alpha_j - n_j\alpha_i)P = -\alpha_i T.$$

The right side is torsion. A nonzero endomorphism is an isogeny and cannot take a nontorsion point to a torsion point. Hence $n_i\alpha_j = n_j\alpha_i$.

Put $g = \gcd(|n_i|, |n_j|)$, $u = n_i/g$, and $v = n_j/g$. There are integers a, b with $au + bv = 1$. Set $\gamma = a\alpha_i + b\alpha_j$; the preceding equality gives $\alpha_i = u\gamma$ and $\alpha_j = v\gamma$. Therefore $R = [g]P + \gamma U$ satisfies $[u]R = O$ and $[v]R = T$. The order of R divides $|u|$, which is prime to N, whereas T has order $N > 1$. This is impossible. Finally, every nonzero α_i is surjective, so the corresponding factor has a pole, and the absence of a zero at that point keeps it a pole of the product. □

Two nontorsion points P_1, P_2 will be called independent over $\mathcal{R}$ modulo torsion if $\alpha P_1 + \beta P_2$ torsion implies $\alpha = \beta = 0$.

Lemma 2. *Suppose P_1, P_2 are independent over $\mathcal{R}$ modulo torsion. If n, m are nonzero integers and α, β are nonzero endomorphisms, the functions $f([n]P_1 + \alpha U)$ and $f([m]P_2 + \beta U)$ have no common pole.*

Proof. A common pole would give $\alpha U = -[n]P_1$ and $\beta U = -[m]P_2$. Commutativity then gives $n\beta P_1 - m\alpha P_2 = O$, contradicting independence. □

2 Excluding abelian translates in an aggregate fiber

Fix nonzero scalars c_1, c_2 and an integer $r \geq 1$. On E^{2r} consider the rational function

$$F(Q_1, \ldots, Q_r, Q'_1, \ldots, Q'_r) = c_1 \prod_{i=1}^{r} f(Q_i) + c_2 \prod_{i=1}^{r} f(Q'_i).$$

Theorem 3. *Let P_1, P_2 be independent over $\mathcal{R}$ modulo torsion. Suppose every n_i, m_i is nonzero and prime to N. Put*

$$q = ([n_1]P_1, \ldots, [n_r]P_1, [m_1]P_2, \ldots, [m_r]P_2).$$

No positive-dimensional abelian translate through q is contained in a level set of F.

Proof. Suppose $q + B$ is such a translate, with B a nonzero abelian subvariety of E^{2r}. There is a nonzero homomorphism $\phi : E \to B$. For example, a nonzero coordinate projection $B \to E$ is surjective; dualizing and composing with an inverse to a polarization up to multiplication produces such a map. Write its coordinates as $(\alpha_1, \ldots, \alpha_r, \beta_1, \ldots, \beta_r)$, not all zero. The restriction of F to $q + \phi(E)$ is

$$c_1 \prod_i f([n_i]P_1 + \alpha_i U) + c_2 \prod_i f([m_i]P_2 + \beta_i U).$$

Within either product a pole survives by Lemma 1. A pole of one product is not a pole of the other by Lemma 2. Since at least one endomorphism is nonzero, at least one product has a pole, and it remains a pole of the sum. This contradicts constancy. □

The same proof permits any finite number of blocks based on points that are pairwise independent over $\mathcal{R}$ modulo torsion. The integers in each block need only be prime to the order of the unique torsion zero of its chosen function.

3 Sums of abscissas without a parity restriction

Give E a Weierstrass equation and write x for its abscissa. This function has its only pole at O, of order two. Let $w_1, \ldots, w_r, w'_1, \ldots, w'_r$ be positive rational numbers and let c_1, c_2 be nonzero algebraic numbers. Define

$$F_+(Q_1, \ldots, Q_r, Q'_1, \ldots, Q'_r) = c_1 \sum_i w_i x(Q_i) + c_2 \sum_i w'_i x(Q'_i).$$

Lemma 4. *Let $P \in E$ have infinite order, let all integers n_i be nonzero, and let $\alpha_i \in \mathcal{R}$. If some $\alpha_i \neq 0$, the function*

$$\sum_i w_i x([n_i]P + \alpha_i U)$$

has a pole and is nonconstant.

Proof. Choose a pole U_0 of a nonconstant summand, say the ith. If another summand, the jth, has a pole there, then $[n_i]P + \alpha_i U_0 = [n_j]P + \alpha_j U_0 = O$. Applying the endomorphisms and subtracting gives $(n_i\alpha_j - n_j\alpha_i)P = O$ and hence $n_i\alpha_j = n_j\alpha_i$.

Let z be a uniformizer at O with $x = z^{-2} + O(z^{-1})$, and use $z(U - U_0)$ locally at U_0. Let λ_j be the nonzero tangent multiplier of α_j at O. The leading coefficient of the jth summand at this pole is $w_j\lambda_j^{-2}$, and the endomorphism identity gives $\lambda_j = (n_j/n_i)\lambda_i$. Consequently the coefficient of z^{-2} in the full sum is

$$\lambda_i^{-2} \sum_{j:\,[n_j]P+\alpha_j U_0=O} w_j (n_i/n_j)^2.$$

The sum is a nonzero positive rational number. The pole therefore cannot cancel. □

Theorem 5. *Suppose P_1, P_2 are independent over $\mathcal{R}$ modulo torsion, and all n_i, m_i are nonzero. No positive-dimensional abelian translate through*

$$q = ([n_i]P_1)_{i=1}^r \times ([m_i]P_2)_{i=1}^r$$

is contained in a level set of F_+.

Proof. Restrict to an elliptic subgroup direction exactly as in Theorem 3. Every nonconstant block sum has a pole by Lemma 4. A pole in one block cannot occur in the other block, because the same argument as Lemma 2 would give a nontrivial endomorphism relation between P_1 and P_2. At least one block is nonconstant, so their weighted sum is nonconstant. □

This result uses no torsion zero of x and imposes no parity condition. Its positivity assumption concerns the fixed rational weights within each block, not the values of the abscissas.

Theorem 6 (Several blocks)**.** *Let $P_1, \dots, P_s$ be nontorsion points on E such that each pair is independent over $\mathcal{R}$ modulo torsion. Fix positive integers r_j, positive rational weights w_{ji}, and nonzero algebraic scalars c_j. On $E^{\sum_j r_j}$ put*

$$F_{\mathrm{sum}}((Q_{ji})) = \sum_{j=1}^{s} c_j \sum_{i=1}^{r_j} w_{ji} x(Q_{ji}).$$

For any tuple $q = ([n_{ji}]P_j)$ with all $n_{ji} \neq 0$, no positive-dimensional abelian translate through q is contained in a fiber of F_{sum}.

There is a product version. For each j suppose $\mathrm{div}(f_j) = m_j[T_j] - m_j[O]$, where T_j has order $N_j > 1$. Replace F_{sum} by

$$F_{\mathrm{product}}((Q_{ji})) = \sum_{j=1}^{s} c_j \prod_{i=1}^{r_j} f_j(Q_{ji}).$$

The same conclusion holds if every n_{ji} is prime to N_j.

Proof. Restrict to a nonzero homomorphism $E \to B$ in any proposed positive-dimensional subgroup B. Some block has a nonconstant coordinate. Within that block a pole survives by Lemma 4, or by Lemma 1 in the product case. A pole cannot occur in two distinct blocks: their two pole equations would contradict the pairwise independence of their base points. The pole therefore survives in the full sum, contradicting constancy. □

The scalars c_j need not be real or positive. Pairwise independence suffices; no independence assumption on the entire collection is required.

4 A linear height comparison

We use the following consequence of Von Buhren's family height theorem [1, Theorem 1.1]. Let A be an abelian variety, S a projective variety, and $V \subseteq A \times S$ an integral closed subvariety. For a fixed subgroup $\Gamma \subseteq A(\overline{\mathbb{Q}})$ of finite rank, there are constants $\mu, \nu \geq 0$ such that

$$\widehat{h}_A(a) \leq \mu h_S(s) + \nu \tag{1}$$

whenever $a \in \Gamma$ lies in the fiber V_s and does not lie on a positive-dimensional abelian translate contained in that fiber. The slope μ can be computed from projective geometric data. The theorem does not provide an effective value for ν.

Corollary 7. *Assume the data of Theorem 3 are algebraic numbers. There are constants $M > 0$ and $M_0 \geq 0$, independent of all the integers n_i, m_i, such that*

$$\sum_{i=1}^{r} (\widehat{h}_E([n_i]P_1) + \widehat{h}_E([m_i]P_2)) \leq Mh(F(q)) + M_0$$

for every tuple in that theorem. Here h is the absolute logarithmic height on $\mathbb{P}^1$. One may choose an effective geometric slope M after fixing compatible height normalizations. No effective M_0 is asserted. The same conclusion holds with F_+ and the hypotheses of Theorem 5, allowing all nonzero labels.

Proof. Let $A = E^{2r}$ and take for $V \subseteq A \times \mathbb{P}^1$ the closure of the graph of F. This is integral and projective. Let Γ be the subgroup generated by P_1 in each of the first r coordinates and by P_2 in each of the last r coordinates. It is finitely generated.

Every indicated q is in the domain of F, since no $[n_i]P_1$ or $[m_i]P_2$ is the identity. If an abelian translate through q were contained in the projective fiber $V_{F(q)}$, it would meet the domain of F at q. Its dense open intersection with that domain would make F constant on the translate. Theorem 3 rules this out. This argument also handles possible boundary components introduced by closing the graph; none through q changes the conclusion.

Apply (1), using a symmetric ample product line bundle on A. Its canonical height is a fixed positive multiple of the sum of the coordinate canonical heights. Rescaling gives the stated inequality. For F_+ the same proof uses Theorem 5 to exclude the exceptional translates. □

For either function of Theorem 6 the same argument gives a uniform inequality

$$\sum_{j,i} \widehat{h}_E([n_{ji}]P_j) \leq Mh(F(q)) + M_0.$$

The subgroup is generated by the finitely many base points placed in their respective coordinates. The constants are fixed before any of the integer labels are chosen.

5 The quadratic-field elliptic curve

Let

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad d' = 3 + \sqrt{2}, \quad E: \ y^2 = x^3 - d^2x,$$

and let σ be the nonidentity automorphism of $K/\mathbb{Q}$. The point used in the rational source is

$$R_0 = (1 + \sqrt{2}, 2\sqrt{2}), \qquad P = 2R_0 = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

For this section its infinite order is the only group property needed; no assertion about the full Mordell–Weil group is required.

Lemma 8. *The displayed point P has infinite order.*

Proof. Substitution verifies $R_0 \in E(K)$, and the usual tangent formula gives the displayed double. At the prime $\mathfrak{p} = (\sqrt{2})$ normalize $v_\mathfrak{p}(\sqrt{2}) = 1$, so $v_\mathfrak{p}(2) = 2$. The displayed coordinates give

$$v_\mathfrak{p}(x(P)) = -6, \qquad v_\mathfrak{p}(y(P)) = -9, \qquad v_\mathfrak{p}(-x(P)/y(P)) = 3.$$

The parameter $z = -x/y$ places P in the formal group of this integral Weierstrass equation. Its multiplication series have integral coefficients and linear term $[m](z) = mz + O(z^2)$. For odd m and $v(z) \geq 3$, the linear term uniquely has least valuation, so $v([m]z) = v(z)$. For multiplication by two, the linear term has valuation $v(z) + 2 < 2v(z)$, and therefore $v([2]z) = v(z) + 2$. No successive multiplication by an odd integer and by a power of two can give zero. Every nonzero integer has this form, proving the claim. □

Lemma 9. *Over a field containing $\sqrt{d}$ and $\sqrt{d'}$, identify E and E^σ with $E_0 : Y^2 = X^3 - X$ by*

$$(X, Y) = \left(\frac{x}{d}, \frac{y}{d\sqrt{d}}\right), \qquad (X', Y') = \left(\frac{x'}{d'}, \frac{y'}{d'\sqrt{d'}}\right).$$

The images P_1, P_2 of P, P^σ are independent over $\mathrm{End}(E_0) = \mathbb{Z}[i]$ modulo torsion.

Proof. The squareclasses of d, d' are independent in $K(i)$. Indeed, a positive element of the real field K that is a square in $K(i)$ is already a square in K: writing a square root as $a + ib$ gives $ab = 0$, and the purely imaginary case would be negative at a real embedding. Neither d nor d' is a square in K, since its norm to $\mathbb{Q}$ is 7. Their product 7 is not a square in K: expanding $(a + b\sqrt{2})^2 = 7$ for rational a, b gives $ab = 0$ and then requires either 7 or $7/2$ to be a rational square.

It follows that $K(i, \sqrt{d}, \sqrt{d'})/K(i)$ has an automorphism fixing $\sqrt{d}$ and negating $\sqrt{d'}$. It fixes P_1, negates P_2, and fixes every endomorphism in $\mathbb{Z}[i]$. If $\alpha P_1 + \beta P_2$ were torsion, applying the automorphism and subtracting would make $2\beta P_2$ torsion. Infinite order forces $\beta = 0$, and then $\alpha = 0$. □

Theorem 10. *For any fixed $r \geq 1$, there are constants $M_+ > 0$ and $M_{+,0} \geq 0$ such that, for all nonzero integers $n_1, \ldots, n_r$,*

$$\sum_{i=1}^{r} \widehat{h}_E([n_i]P) \leq M_+ h\left(\frac{1}{2}\,\mathrm{Tr}_{K/\mathbb{Q}} \sum_{i=1}^{r} x([n_i]P)\right) + M_{+,0}. \tag{2}$$

There are also constants $M_\times > 0$, $M_{\times,0} \geq 0$ such that, when all the labels are odd,

$$\sum_{i=1}^{r} \widehat{h}_E([n_i]P) \leq M_\times h\left(\frac{1}{2}\,\mathrm{Tr}_{K/\mathbb{Q}} \prod_{i=1}^{r} x([n_i]P)\right) + M_{\times,0}. \tag{3}$$

Both slopes can be chosen effectively from the fixed geometry; the additive constants are not determined effectively here.

Proof. On E_0, the coordinate X has divisor $2[(0,0)] - 2[O]$. Its zero is a nonzero point of order two, so the index condition in Theorem 3 is exactly oddness. Under the isomorphisms of Lemma 9, the rational function in the height on the right of (3) is

$$F = \frac{d^r}{2} \prod_{i=1}^{r} X_i + \frac{(d')^r}{2} \prod_{i=1}^{r} X_i'.$$

For the sum, the corresponding function is $F_+ = (d/2) \sum_i X_i + (d'/2) \sum_i X_i'$ and Theorem 5 applies without parity restrictions. Apply Corollary 7 to both functions with $m_i = n_i$. Canonical heights are unchanged by these pointed isomorphisms, and conjugation preserves absolute canonical height. The coordinate height sum there is twice the sum in (3); divide by two. □

Notice that the inequality is linear. Finiteness of each individual fiber alone would not give this conclusion. The additive constant is the remaining effectiveness issue; it is not an additional mathematical conjecture.

6 A trace-height consequence over number fields

Corollary 11. *Let K/F be a finite extension of number fields, let E/K be an elliptic curve, and let $P \in E(K)$ have infinite order. For every embedding $\sigma : K \hookrightarrow \overline{F}$ over F, suppose there is a pointed isomorphism $\phi_\sigma : E^\sigma \to E_0$ to a fixed elliptic curve over $\overline{F}$. Assume that the points $P_\sigma = \phi_\sigma(P^\sigma)$ are pairwise independent over* $\mathrm{End}(E_0)$ *modulo torsion.*

For every fixed positive integer r there are constants $M > 0$, $M_0 \geq 0$ such that, for all nonzero integer labels,

$$\sum_{i=1}^{r} \widehat{h}_E([n_i]P) \leq Mh\left(\mathrm{Tr}_{K/F} \sum_{i=1}^{r} x([n_i]P)\right) + M_0.$$

The slope can be chosen effectively from the fixed geometric data; the additive constant has the same ineffective status as in Corollary 7.

Proof. Under a pointed isomorphism of Weierstrass models the abscissa transforms as $x^\sigma = a_\sigma x_0 + b_\sigma$, with $a_\sigma \neq 0$. The function in the height is therefore

$$\sum_{\sigma} a_\sigma \sum_{i} x_0([n_i]P_\sigma) + r \sum_{\sigma} b_\sigma.$$

Theorem 6 excludes constant translates through the indicated points for the nonconstant part. Adding the fixed scalar does not change that conclusion. The graph-family argument gives a linear bound for the sum of the canonical heights in all the embedding blocks. Absolute canonical height is unchanged by conjugation and pointed isomorphism, so this sum is $[K : F]$ times the left side of the displayed inequality. Divide by $[K : F]$. □

This corollary is a height comparison under explicit independence hypotheses. It does not assert a Diophantine definition of the ring of integers of K.

7 A conditional existence result with twenty-seven variables

We specify precisely how the height theorem applies to the explicit rational source in [2]. We retain its group, contact-height, pole, and simultaneous prime-pattern premises. Its contact exponent is $\eta = 19$, its comparison orders are 39 and 10, and its scalar order is 20. These premises and the same finite-support completeness argument are the only arithmetic inputs in the following consequence.

Write the four root labels as (n, a, t, b), all nonzero integers, and put

$$B = \max(|n|, |a|, |t|, |b|), \quad c = |a - 1|, \quad \Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1,$$

and use the positive rational number

$$F = \mathrm{Tr}_{K/\mathbb{Q}}(x([n]P) + x([a]P) + x([t]P) + x([b]P)).$$

Positivity follows because the source points are nonzero doubles and their abscissas are positive at both real embeddings. In particular $F \neq 0$. The sum uses only additions after the four abscissas have been computed. Replacing the half trace in Theorem 10 by the full trace changes the height by at most $\log 2$, absorbed into its additive constant. The product theorem

gives an independent height comparison when all labels are odd; the construction here uses the sum.

Use only the two comparison scales 1 and F^2. The source's contact-height inequality for these scales gives

$$h(F) \le (H + \log 3)|N|^{19/k} \tag{4}$$

whenever their local certificates give the required k-fold contact divisibility for the nonzero integer N. Indeed, the unscaled and scaled arguments bound the heights of λ and $F^2\lambda$ by the same right side, so $2h(F) = h(F^2) \le h(\lambda) + h(F^2\lambda)$. Theorem 10, with $r = 4$, then implies

$$B^2 \le M_1 h(F) + M_2 \tag{5}$$

for fixed $M_1 > 0$, $M_2 \ge 1$. This follows directly from $\widehat{h}_E([j]P) = j^2\widehat{h}_E(P)$ and $\widehat{h}_E(P) > 0$.

Theorem 12. *Under the stated arithmetic premises, there exists a polynomial over $\mathbb{Z}$ in twenty-seven total variables, including its input variable, of total degree at most 2^{158}, whose rational zero set projects exactly to $\mathbb{Z}$. The proof does not determine all of its coefficients.*

Proof. For any positive odd integer L, the scalar normalizer can be chosen as

$$R_L(X) = \frac{102^2\ell^L r(X)}{(1 + C_L r(X))^2}, \qquad \ell = 76249, \qquad C_L = 17(\ell^{(L+1)/2} + 1),$$

with $\beta = 216X^4 + 450459$ and $r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1)$. The finite-field certificate in [4] proves that r and $1 + 17r$ have no zero modulo ℓ. Thus

$$v_\ell(R_L) = L \quad (v_\ell X \ge 0), \qquad v_\ell(R_L) = L - 20v_\ell X \quad (v_\ell X < 0).$$

Also $0 < R_L < 1$, since $C_L^2 r(0) > 17^2\ell^{L+1} r(0) > 102^2\ell^L$. All other scalar local conditions and square-denominator conditions are unchanged. The fixed-prime extraction argument therefore gives

$$v_\ell(a - 1) \ge 19L + 1, \qquad c \ge \ell^{19L+1}.$$

If $\Delta \ne 0$, local comparison and (4) with $k = 39$ give, using $|\Delta| < 5B^4$,

$$B^2 \le M_1(H + \log 3)(5B^4)^{19/39} + M_2.$$

The exponent $76/39$ is strictly smaller than two. Hence this inequality fails for all sufficiently large B. Since $B \ge |a| \ge c - 1$, choosing the fixed odd L sufficiently large excludes $\Delta \ne 0$ for every solution.

Now $\Delta = 0$, and the elementary integer Lucas growth argument gives $\log B > c - 4$ once $|a| > 7$. If $\beta \ne n$, write N for the absolute numerator of $\beta - n$ in lowest terms. The source gives $N \le 2e^{h(\beta)}B$ and contact divisibility of order ten. Equations (4) and (5) give

$$B^2 \le M_1(H + \log 3)(2e^{h(\beta)}B)^{19/10} + M_2.$$

For $B^2 > 2M_2$, taking logarithms yields $\log B \le 19h(\beta) + M_3$ for a fixed constant M_3. The scalar estimate remains $h(\beta) \le Hc^{19/20}$. It follows that

$$c - 4 < 19Hc^{19/20} + M_3,$$

which fails for all sufficiently large c. Increase the same fixed odd L if necessary. We conclude $\beta = n \in \mathbb{Z}$, and the source input map then gives $X \in \mathbb{Z}$.

Conversely, for each integer X the unchanged completeness argument prepares the finite support of $R_L(X)$ and applies the simultaneous prime-pattern premise to the two remaining

scales. Removing a comparison scale removes requirements; the source's finite-avoidance argument still applies. It therefore supplies all rational witnesses.

The retained coordinates are the input, eighteen coordinates for nine elliptic point charts, one slope, four parameters for the two comparison poles, and two parameters for the scalar pole: $1+18+1+4+2 = 26$. The finite-algebra elimination theorem of [3] adds one coordinate, giving twenty-seven. Its rational projection is unchanged by that elimination. Clearing the fixed rational coefficients gives a polynomial over $\mathbb{Z}$.

All constants used to choose L exist by the proved height theorem. Their existence proves the polynomial's existence. The ineffective additive constant M_2 prevents this proof from specifying a certified numerical value of L.

The direct point charts and the single cubic change of basis for their square roots are proved in [6, 7]. The resulting finite-algebra specification [5] has nine quartic point generators and three quadratic pole generators. Its maximum rank is $4^9 \cdot 2^3 = 2^{21}$, with eight source cases. Exact degree arithmetic gives the majorant

$$232075682770882905150152990100588354340946706432 < 2^{158}.$$

The exponent L enters only integer coefficients. Every degree computation and the same majorant are therefore uniform in the fixed odd $L \geq 3$, including every sufficiently large member used above. This proves the degree assertion without an effective value of L. □

The formal logarithm orders remain 39, 10, and 20; only a fixed scalar coefficient is enlarged. Thus the ineffectiveness does not conceal an extra coordinate or a degree that grows with L: L occurs solely in coefficients. The explicit twenty-nine-variable construction remains a different deliverable, with all constants given.

References

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[2] *An explicit polynomial in 29 variables over the rationals*, project manuscript, 8 November 2026. Project repository; source file `few_variables_20261108_v8.tex`.

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[6] *Square abscissas and direct charts for the elliptic subgroup*, companion manuscript, 8 November 2026. Source file `square_abscissa_charts_20261108_v9.tex`.

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